SIMATS ENGINEERING

ASSESSMENT TOOL 2 : Concept Mapping

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Total Marks: 50

Dave's Taxonomy: Imitation (Level 1)
Question Count: 4

Code of Conduct

I Mohamed Saif.A / 192521171 (Name / Reg No)

certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Assessment Tasks

Concept Mapping Tasks

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.
2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.
3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.
4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.

Expected concept nodes:

- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

3.

Service models

Iaas

=> Physical data centre

=> Networking & Storage Infrastructure

=> Security & availability

=> manage os, application & data

paas

=> Infrastructure & hardware

=> operating system

=> Runtime environment & middleware

=> eg: Azure

Saas

=> Application Software

=> Data Storage management

=> manage own data

=> usage policies

eg: Salesforce

Xaas

=> Refer to any Cloud Service

=> provide specialized Cloud Service

=> ensure security performance

=> handle update security

EVOLUTION OF CLOUD COMPUTING

Hardware Evolution

=> From mainframe (1960's - 1970's) to personal computer

=> Rise virtualization and multicore-processor (2000's)

=> high performance services SSDS, CWC & scalable

Distributed System

=> From single to machine system

=> peer to peer grid computing

=> Scalable

Service delivery

=> Utility computing

=> Emergence of IaaS, PaaS, SaaS

=> Serverless & edge computing

Internet software evolution

=> Static web page

=> Dynamic Web

=> Cloud Native

Service models

IaaS

4)

Cloud Computing & Virtualization

Virtualization

=> enable multitenancy
=> provide on demand
=> Reduce hardware cost
=> Isolation & security

Web Service

=> Connects application & service across the cloud
=> Supports hybrid and multicloud environment
Eg: HTTP, REST

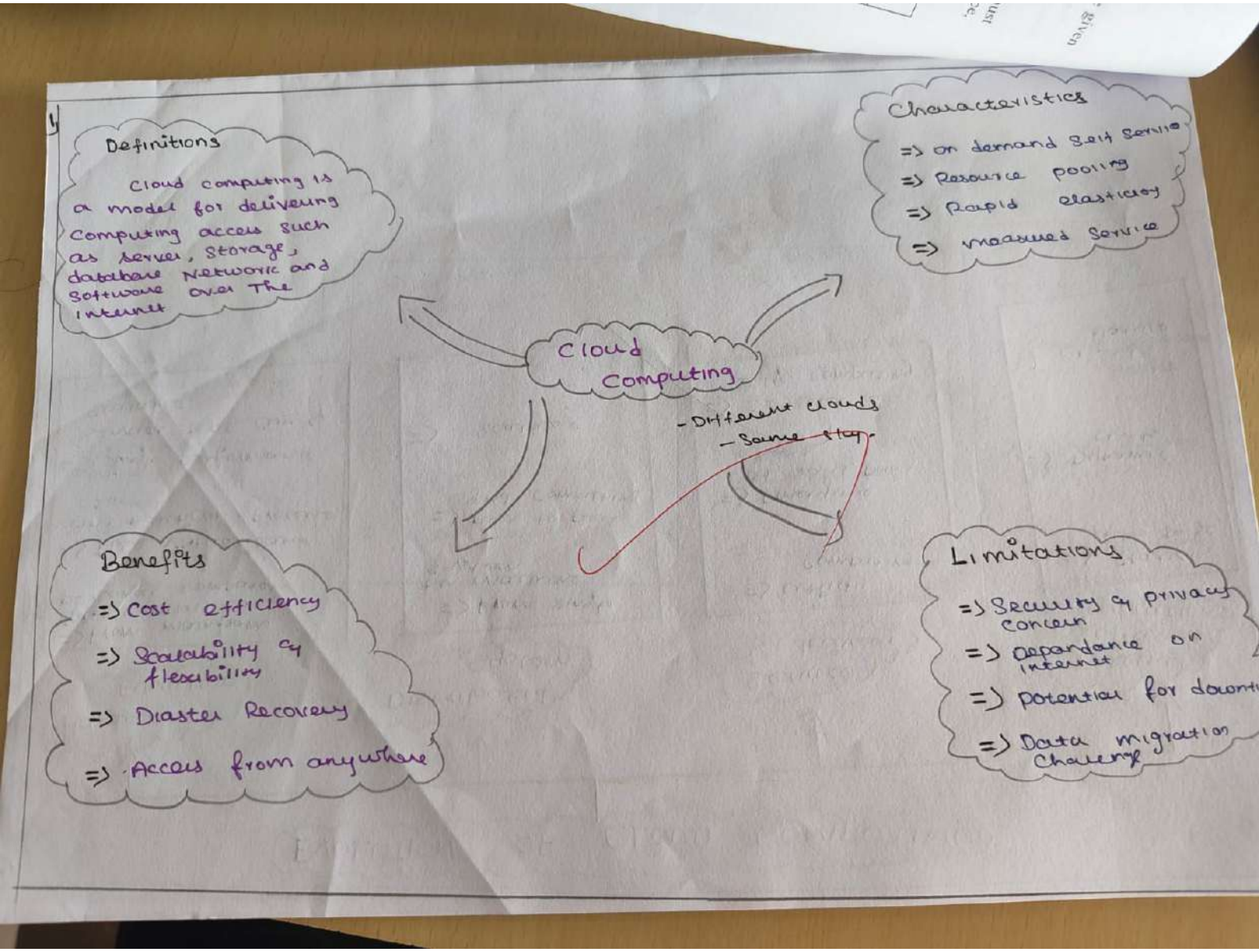
Enable cloud Deployment

=> Virtualization abstract hardware, allowing clouds provider to run multiple instance
=> Together they simplify setup, reduce complexity & speedup

Enable Scalability

=> Virtualization allow dynamic provisioning and live migration of virtual machine
=> web service enable seamless scaling through API load balance

13/10/25


Definitions
Cloud computing is a model for delivering computing access such as server, storage, database Network and software over The internet
Characteristics
=> on demand Self Servic
=> Resource pooling
=> Rapid elasticity
=> measured service
Cloud Computing
- Different clouds
- Same step
Benefits
=> Cost efficiency
=> Scalability & flexibility
=> Diaster Recovery
=> Access from anywhere
Limitations
=> Security & privacy concern
=> Depandance on internet
=> potential for downt
=> Data migration Challeng